

Diabetes Disease Classification Using Machine Learning Techniques

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Abstract—This paper presents a machine learning framework for diabetes classification using patient health and lifestyle data. The dataset includes features such as gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level, and blood glucose level to predict whether a patient has diabetes. Exploratory Data Analysis (EDA) and preprocessing techniques, including handling missing values, encoding categorical features, and feature scaling, were applied to prepare the data. Principal Component Analysis (PCA) was also investigated for dimensionality reduction and compared with models trained without PCA. Several classification algorithms were implemented, including Logistic Regression, MLP, SVM, Naïve Bayes, and ensemble methods. Model performance was evaluated using accuracy, precision, recall, F1-score, confusion matrices, and learning curves to analyze model behavior and compare results. The proposed approach provides an effective framework for accurate diabetes prediction using machine learning techniques.

Index Terms—Machine Learning, Regression, Price Prediction, Linear Regression, Decision Trees, Cross-Validation

I. INTRODUCTION

Diabetes is one of the most common chronic diseases worldwide and can lead to serious health complications if not detected and managed early. Accurate prediction of diabetes using medical and lifestyle information can assist healthcare professionals in improving diagnosis and decision-making processes. With the rapid growth of healthcare data, machine learning techniques have become powerful tools for analyzing patient information and building predictive models for disease classification.

This paper presents a machine learning framework for diabetes disease classification using patient health data, including gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level, and blood glucose level. The study applies Exploratory Data Analysis (EDA) and preprocessing techniques such as feature scaling and categorical encoding to prepare the dataset for modeling. In addition, Principal Component Analysis (PCA) is investigated to evaluate the effect of dimensionality reduction on classification performance.

Several machine learning algorithms are implemented and compared, including Logistic Regression, Multi-Layer Percep-

tron (MLP), Support Vector Machine (SVM), Naïve Bayes, and ensemble learning methods. Hyperparameter optimization techniques are used to improve model performance. The models are evaluated using performance metrics including accuracy, precision, recall, F1-score, confusion matrices, and learning curves. The objective of this work is to identify the most effective classification model for diabetes prediction and analyze the impact of PCA on model accuracy and generalization.

II. EDA

Exploratory Data Analysis (EDA) was conducted to understand the structure of the dataset, examine feature distributions, detect potential outliers, and analyze relationships between the input variables and the target variable. This stage is essential for identifying meaningful patterns in the data and guiding subsequent preprocessing and modeling steps.

The dataset consists of multiple medical and lifestyle-related features describing patient health conditions, including gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level, and blood glucose level. **The target variable in this study is Diabetes, which indicates whether a patient is diabetic or non-diabetic.**

The analysis includes both categorical and numerical features, each requiring different visualization and analytical techniques.

A. Class Distribution

The dataset is imbalanced, where non-diabetic patients represent the majority of the data.

Non-diabetic (0): 91,500 samples (91.5%)
Diabetic (1): 8,500 samples (8.5%)

This imbalance indicates that accuracy alone is not sufficient for evaluation, since a model may achieve high accuracy while performing poorly on diabetic cases. Therefore, additional metrics such as Precision, Recall, F1-score, and Confusion Matrix were used.

B. Gender Distribution

The dataset contains more female patients than male patients, while the “Other” category appears very rarely. This suggests that gender distribution is relatively balanced between males and females, with no major skew affecting the dataset.

C. Hypertension and Heart Disease

Most patients do not suffer from hypertension or heart disease. However, diabetic patients appear more frequently among individuals with these conditions, indicating a potential relationship between chronic diseases and diabetes risk.

D. Smoking History

The majority of patients fall under the “never” and “No Info” smoking categories. Patients with a history of smoking, especially former smokers, show relatively higher diabetes occurrence compared to other groups.

E. Age Distribution

The age distribution covers a wide range, with diabetic patients generally belonging to older age groups. The boxplot shows that the median age of diabetic patients is significantly higher than non-diabetic patients, suggesting age is an important predictive feature.

F. BMI Distribution

BMI values are right-skewed with several outliers. Diabetic patients tend to have higher BMI values compared to non-diabetic patients, indicating a correlation between obesity and diabetes risk.

G. HbA1c Level and Blood Glucose Level

Both HbA1c level and blood glucose level show noticeable differences between diabetic and non-diabetic patients. Diabetic patients generally have significantly higher values, making these two features among the strongest indicators for diabetes prediction.

H. Correlation Analysis

The correlation heatmap shows weak to moderate correlations among numerical features. The strongest correlation appears between age and BMI (0.34), while HbA1c and blood glucose level show a smaller positive correlation. No severe multicollinearity was observed, which supports the suitability of the features for machine learning models.

III. DATA PREPROCESSING

Data preprocessing is a critical step in machine learning pipelines, as the quality of the input data directly affects model performance. The goal of this stage is to ensure that the dataset is clean, consistent, and suitable for classification tasks. Several preprocessing steps were performed, including missing value detection, duplicate removal, handling inconsistent categorical values, feature encoding, feature scaling, and outlier analysis.

These preprocessing techniques help ensure that the dataset accurately represents patient health information before applying machine learning algorithms for diabetes prediction.

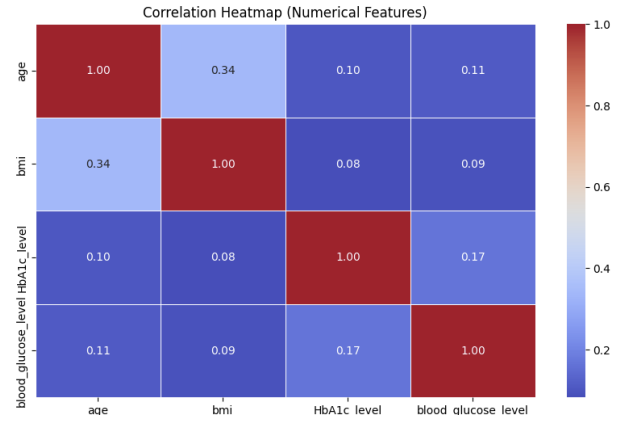


Fig. 1. Correlation between the different (Numerical Attributes)

A. Missing Value Detection

The first preprocessing step involved examining the dataset for missing values. Missing data can negatively affect model training by introducing incomplete information or bias.

To identify missing values, a column-wise null value count was computed using the `isnull()` and `sum()` functions.

The analysis revealed that the dataset does not contain any missing values. Therefore, no imputation or removal strategies were required for handling null entries. This ensures that all records can be used during model training without the need for additional data completion techniques.

B. Handling Non-Informative Features

Certain features may provide little or no meaningful contribution to the classification process. Features with completely identical values or features containing highly unique values may not help machine learning models identify useful patterns related to diabetes prediction.

To analyze feature usefulness, the number of unique values in each column was computed using the `nunique()` function. This step helps identify:

- Features with a single repeated value across all records.
- Features with extremely high uniqueness that may behave similarly to identifiers.

The analysis showed that the dataset did not contain explicit identifier columns such as patient IDs. Most features contained meaningful variations that could contribute to diabetes classification. Features such as gender, smoking history, hypertension, heart disease, BMI, HbA1c level, and blood glucose level demonstrated useful variability and were therefore retained for model training.

This analysis ensures that the dataset contains informative attributes only, reducing unnecessary complexity and improving the effectiveness of machine learning algorithms in learning patterns associated with diabetes prediction.

C. Handling Duplicate Values

Duplicate records were examined to ensure that repeated patient entries did not negatively affect the learning process.

Duplicate samples can bias machine learning models and distort statistical analysis by overrepresenting certain patterns within the dataset.

The analysis showed that the dataset originally contained 3,854 duplicate rows out of 100,000 total records. These duplicate entries were removed, reducing the dataset size to 96,146 records. After preprocessing, the dataset was verified again to confirm that no duplicate rows remained.

In addition, the dataset was checked for missing values, and the analysis showed that no null values were present. Therefore, no rows needed to be removed due to missing data.

The categorical feature `smoking_history` contained the category “No Info”, which represents unclear or unavailable smoking information. To improve data consistency and reduce ambiguity, these values were replaced with the mode (most frequent valid category), which was “never”. After imputation, the majority of patients belonged to the “never” smoking category, followed by “former” and “current” smokers.

Handling duplicate and inconsistent values improves dataset quality and ensures that the machine learning models are trained on reliable and meaningful patient information.

D. Handling Garbage Values

Garbage values refer to incorrect, inconsistent, or unexpected data entries that may negatively affect model performance and reduce data quality. To identify such values, categorical features were manually inspected by examining the frequency of each category within the dataset.

The analysis showed that the `gender` feature contained three categories: “Female”, “Male”, and “Other”. The “Other” category appeared only 18 times out of 96,146 records, representing a very small portion of the dataset. Since the values were still valid and not erroneous, they were retained rather than removed.

For the `smoking_history` feature, the dataset contained meaningful categories including “never”, “former”, “current”, “not current”, and “ever”. After replacing the ambiguous “No Info” entries during preprocessing, the remaining categories were consistent and medically meaningful, indicating that no invalid or unexpected categorical values remained in the dataset.

Inspecting categorical features for garbage values helps improve dataset reliability and ensures that the machine learning models are trained using clean and meaningful patient information.

E. Outlier Detection

Outliers are observations that significantly deviate from the majority of the dataset and may affect model training and evaluation. To investigate potential outliers, descriptive statistics and boxplots were generated for all numerical features.

Several indicators were considered when identifying possible outliers:

- A large difference between the upper quartile and the maximum value, indicating the presence of high-end outliers

- A noticeable difference between the mean and median values, suggesting skewed feature distributions
- Extreme values appearing outside the whiskers in boxplot visualizations

The analysis showed that the BMI feature contained the most noticeable outliers, while HbA1c level and blood glucose level contained a smaller number of upper-range outliers, which were neglected as outliers not too extreme and can be actual real-world cases.

However, these values were considered medically realistic rather than erroneous data. For example, very high BMI values may correspond to obesity cases, while elevated HbA1c and blood glucose levels may represent patients with severe diabetic conditions.

Therefore, the outliers were handled carefully instead of being completely removed. BMI values were capped using the Interquartile Range (IQR) method to reduce the influence of extreme observations while preserving realistic medical cases. Other numerical features were retained because their outliers were limited and medically meaningful.

Removing all extreme values could reduce the model’s ability to learn important patterns associated with high-risk diabetic patients.

These refinements ensured that the dataset was both memory-efficient and interpretable, avoiding feature explosion while preserving meaningful brand-level information. This preprocessing stage provided a robust foundation for subsequent encoding and model training.

IV. MODEL EVALUATION

To ensure consistent and reproducible comparison across different algorithms, a generalized evaluation function was implemented. This function integrates with the preprocessing pipeline described in Section E and applies K-Fold cross-validation to compute both training and validation metrics for each fold. By centralizing the evaluation process, redundant code was avoided and results could be systematically compared.

A. Evaluation Function

The evaluation function accepts a model instance and configurable options (e.g., polynomial expansion, feature scaling). For each fold, the following steps are performed:

- Train the model on the training subset.
- Generate predictions for both training and validation subsets.
- Compute performance metrics: coefficient of determination (R^2 , measures how much variance is explained), root mean squared error (RMSE, shows the typical size of prediction errors), and mean absolute error (MAE, reflects the average difference between predicted and actual values).
- Store fold-specific results in a tabular format for inspection.

After all folds are processed, the function reports average training and validation scores, providing a concise summary

of model performance. This design ensures that each model can be evaluated under identical conditions, facilitating fair comparison.

B. Linear Regression

The baseline linear regression model achieved strong training performance with an average R^2 of 0.97, but validation performance dropped to 0.77. This gap indicates mild overfitting: the model explains most of the variance in training data but generalizes less effectively to unseen folds. The average validation RMSE of 328.5 and MAE of 196.6 suggest that while predictions are reasonable, errors remain relatively large for premium or atypical laptops. Overall, linear regression provides a useful benchmark but struggles with generalization.

C. Random Forest

Random Forest was implemented as a bagging ensemble learning algorithm that combines multiple decision trees to improve classification stability and reduce overfitting. The model was trained using the SMOTE-balanced training dataset after preprocessing and scaling.

Hyperparameter tuning was applied using GridSearchCV to optimize the following parameters:

- Number of estimators
- Maximum tree depth
- Minimum samples split
- Minimum samples leaf

Learning curves were also generated to analyze the model's convergence behavior and detect possible overfitting or underfitting.

Results: The Random Forest model achieved excellent training performance with an accuracy of 99.92%, precision of 99.92%, recall of 99.88%, and F1-score of 99.90%. Validation performance remained strong with an accuracy of 95.53% and F1-score of 73.88%, indicating good generalization despite a slight decrease compared to the training results.

When it comes to results, On the testing dataset, which is our main concern the model achieved an accuracy of 96.01%, precision of 78.52%, recall of 75.31%, and F1-score of 76.89%. These results demonstrate that Random Forest provided strong predictive capability and handled the diabetes classification task effectively while maintaining stable performance across unseen data.

D. AdaBoost

AdaBoost was implemented as a boosting ensemble learning algorithm. The model sequentially trains weak learners while increasing focus on samples that were previously misclassified. This adaptive learning behavior allows the model to improve classification performance on difficult cases.

Hyperparameter tuning was performed using GridSearchCV to optimize:

- Number of estimators
- Learning rate

Learning curves were plotted to evaluate the relationship between training and validation accuracy as the training size increased.

Results: The AdaBoost model achieved strong training performance with an accuracy of 93.08%, precision of 91.75%, recall of 91.40%, and F1-score of 91.58%.

Validation results showed an accuracy of 92.70%, precision of 55.84%, recall of 82.64%, and F1-score of 66.65%. The higher recall compared to precision indicates that AdaBoost was effective in identifying diabetic cases while producing a moderate number of false positives.

On the testing dataset, the model achieved an accuracy of 93.37%, precision of 58.60%, recall of 84.67%, and F1-score of 69.26%. These results suggest that AdaBoost demonstrated strong sensitivity toward diabetic patients and maintained good generalization capability across unseen data.

V. MULTI-LAYER PERCEPTRON (MLP) CLASSIFIER

A. Model Overview

The Multi-Layer Perceptron (MLP) is a feedforward Artificial Neural Network (ANN) consisting of one or more hidden layers between the input and output layers. Unlike the Single Layer Perceptron (SLP), which can only solve linearly separable problems, the MLP is capable of learning complex non-linear decision boundaries through the Backpropagation algorithm combined with Gradient Descent optimization [?].

For binary classification, the MLP minimizes the Binary Cross-Entropy loss function defined as:

$$J(\Theta) = -\frac{1}{m} \sum_{e=1}^m [Y^e \cdot \log(a^e) + (1 - Y^e) \cdot \log(1 - a^e)] \quad (1)$$

where m is the number of training examples, Y^e is the true label, and a^e is the predicted output probability.

B. Hyperparameter Tuning

Hyperparameter optimization was performed using GridSearchCV with 3-fold cross-validation, scored on F1 due to class imbalance. The search space is defined in Table I.

TABLE I
MLP HYPERPARAMETER SEARCH SPACE

Hyperparameter	Values Searched
Hidden Layer Sizes	(64,), (128,), (64,32), (128,64)
Activation Function	Logistic (Sigmoid)
Alpha (L2 Regularization)	0.0001, 0.001
Learning Rate	0.001, 0.0001
Max Iterations	500

The logistic (sigmoid) activation function was selected as it aligns with the course material [?], defined as:

$$g(z) = \frac{1}{1 + e^{-\theta^T x}} \quad (2)$$

Early stopping with `n_iter_no_change=20` and L2 regularization were applied to prevent overfitting, consistent with the regularization concepts introduced in the course [?].

C. Results

TABLE II
MLP PERFORMANCE METRICS

Metric	Validation Set	Test Set
Accuracy	0.9129	0.9181
Precision	0.50	0.5213
Sensitivity (Recall)	0.86	0.8758
F1 Score	0.64	0.6536

The MLP achieved a test accuracy of 91.81%. The model demonstrates high sensitivity (recall) of 87.58%, meaning it correctly identifies the majority of actual diabetic patients — which is the most clinically important metric in disease classification tasks, as missing a diabetic patient (false negative) is more costly than a false alarm. The relatively lower precision of 52.13% indicates that some non-diabetic patients are flagged as diabetic, which is an acceptable trade-off given the class imbalance in the dataset (91% non-diabetic vs 9% diabetic).

D. Learning Curve Analysis

Fig. 2 illustrates the MLP learning curves across approximately 72 training iterations.

The training loss (Binary Cross-Entropy) begins at approximately 0.35 and decreases smoothly to approximately 0.175, demonstrating stable convergence. The validation loss remains consistently low at approximately 0.09, indicating strong generalization to unseen data.

The validation accuracy begins at approximately 0.889, climbs steadily to 0.912 by iteration 30, and stabilizes between 0.910 and 0.915 for the remainder of training. No significant overfitting is observed as validation accuracy does not degrade over iterations. The gap between training and validation loss is partially attributed to the presence of SMOTE-generated synthetic samples in the training set, which introduce complexity not present in the real validation data.

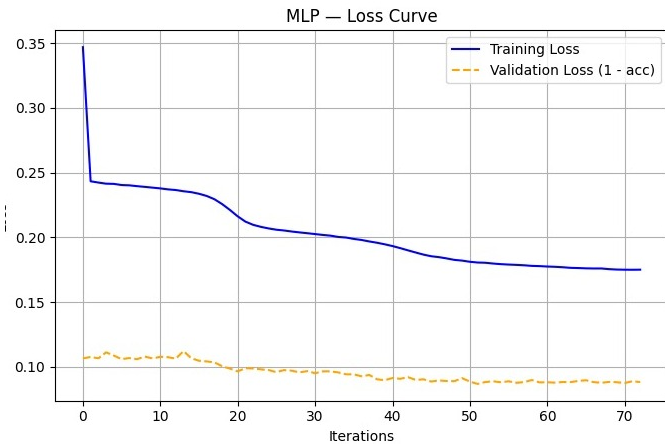


Fig. 2. MLP Learning Curves — and Validation Accuracy vs Iterations

E. Gaussian Naive Bayes (GNB)

Gaussian Naive Bayes is a probabilistic machine learning classification algorithm based on Bayes' Theorem. It predicts

the class of a sample by calculating the probability that the sample belongs to each class and selecting the class with the highest probability.

The model is called “Naive” because it assumes that all input features are statistically independent from each other, which simplifies the computation process and makes the model very fast and efficient.

Since the dataset contains continuous numerical features, the Gaussian Naive Bayes version was used. This variant assumes that each feature follows a normal (Gaussian) distribution within each class.

The probability is calculated using Bayes' theorem:

$$P(C | X) = \frac{P(X | C) \times P(C)}{P(X)}$$

The Gaussian Naive Bayes model demonstrated extremely high sensitivity in detecting diabetic patients, achieving a recall of 96.86 percentage, which makes it effective for minimizing missed positive cases. However, the model suffered from low precision and moderate overall accuracy due to a large number of false positive predictions.

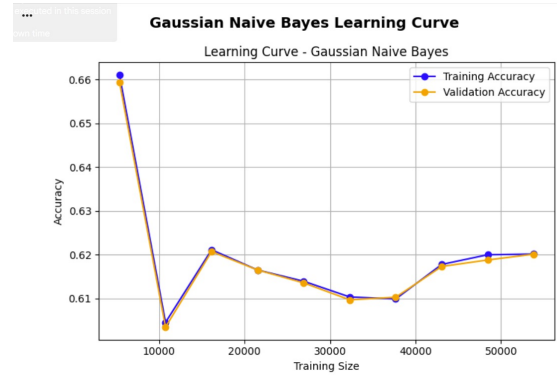


Fig. 4. GNB - Learning Curve

F. Logistic Regression

Logistic Regression was implemented as the baseline classification model for diabetes prediction. The model estimates the probability of diabetes occurrence using a weighted combination of the input features followed by the sigmoid activation function. Due to its simplicity, efficiency, and interpretability, Logistic Regression serves as a strong baseline for evaluating more advanced machine learning models.

Hyperparameter tuning was performed using GridSearchCV with 5-fold cross-validation to optimize:

- Regularization strength (C)
- Penalty type
- Solver type
- Class weighting strategy

The best hyperparameters were:

- $C = 0.01$
- Penalty = L2
- Solver = liblinear

- Class Weight = balanced

The tuned model achieved the best cross-validation F1-score of 86.72%.

1) *Results*: The Logistic Regression model achieved strong training performance with an accuracy of 88.80%, precision of 84.67%, recall of 88.88%, and F1-score of 86.72%.

Validation performance remained stable with an accuracy of 88.00%, precision of 41.48%, recall of 87.59%, and F1-score of 56.30%. The high recall indicates that the model was highly effective at identifying diabetic patients, although this came with an increased number of false positives.

On the testing dataset, the model achieved an accuracy of 88.55%, precision of 42.76%, recall of 88.05%, and F1-score of 57.57%. These results demonstrate that Logistic Regression maintained strong sensitivity toward diabetic cases while preserving good overall generalization capability across unseen data.

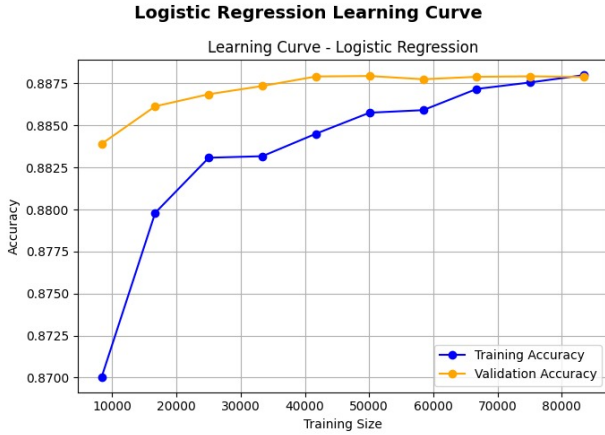


Fig. 5. Logistic Regression Learning Curve

Learning curve analysis showed healthy and stable model behavior. The training and validation curves converged smoothly as the training size increased, while the gap between them remained very small. Training accuracy gradually improved over time, and validation accuracy stayed consistently close to the training performance. This behavior indicates low overfitting, good generalization, and stable learning performance throughout the training process.

G. Support Vector Machine (SVM)

Support Vector Machine (SVM) was implemented for diabetes prediction to identify the optimal decision boundary between diabetic and non-diabetic patients. Different kernels, including Linear, RBF, and Polynomial, were evaluated.

Hyperparameter tuning was performed using Randomized Search Cross-Validation and Particle Swarm Optimization (PSO). The optimization process focused on selecting the best kernel and tuning the parameters C , γ , and polynomial degree using a generic PSO fitness function based on cross-validation accuracy.

Table III compares the best kernel performances.

TABLE III
SVM KERNEL COMPARISON

Kernel	Best Accuracy
RBF	96.706%
Polynomial	96.44%

Since the RBF kernel achieved the highest accuracy, it was selected as the final SVM model using $C = 100$ and $\gamma = 0.1$.

1) *Results Evaluation*: The final RBF SVM model achieved high accuracy and precision across the training, validation, and testing datasets, indicating strong classification capability and good generalization performance. The model maintained stable results on unseen data, demonstrating low overfitting.

Although recall values were lower than precision, the model still achieved balanced F1-scores, showing effective detection of diabetic cases while minimizing false positives. Overall, the RBF SVM provided reliable and consistent performance for diabetes prediction.

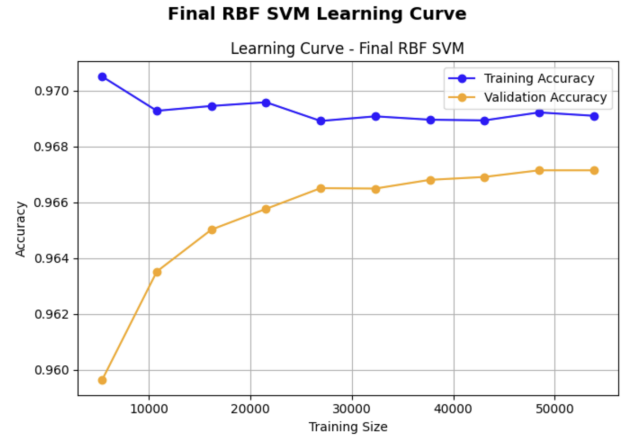


Fig. 6. Final RBF SVM Learning Curve

2) *Learning Curve Evaluation*: Figure 6 shows the learning curve of the final RBF SVM model. The training and validation accuracies remain very close as the training size increases, indicating strong generalization performance and minimal overfitting.

The validation accuracy steadily improves with additional training data before stabilizing, while the training accuracy remains consistently high. The small gap between both curves demonstrates that the model achieved good bias-variance balance and stable predictive performance.

H. Final Model Comparison

Multiple machine learning models were evaluated for diabetes prediction, including Logistic Regression, Gaussian Naive Bayes, MLP, AdaBoost, Random Forest, XGBoost, and the final RBF SVM model. The models were compared using accuracy, precision, recall, and F1-score across the training, validation, and testing datasets.

Among all evaluation metrics, the F1-score was considered the most important metric because the dataset is imbalanced.

In imbalanced classification problems, accuracy alone can be misleading since a model may achieve high accuracy while failing to correctly identify minority diabetic cases. The F1-score provides a balanced evaluation by combining both precision and recall, making it more suitable for assessing diabetes prediction performance.

Table IV summarizes the testing F1-scores of the evaluated models.

TABLE IV
FINAL MODEL COMPARISON USING TESTING F1-SCORE

Model	Testing F1-Score
XGBoost	0.8010
Final RBF SVM	0.7917
Random Forest	0.7689
AdaBoost	0.6926
MLP	0.6536
Logistic Regression	0.5757
Gaussian Naive Bayes	0.3117

XGBoost achieved the highest testing F1-score, indicating the best overall balance between precision and recall. The final RBF SVM model also demonstrated strong performance with high accuracy and precision, while Random Forest achieved competitive results but showed signs of overfitting due to the large gap between training and validation performance.

Overall, XGBoost was selected as the final best-performing model for diabetes prediction because it achieved the strongest balance between correctly identifying diabetic patients and minimizing false predictions.

Since XGBoost achieved the highest testing F1-score, it was selected as the best-performing model. Therefore, PCA was applied to the XGBoost classifier to further evaluate the effect of dimensionality reduction on model performance and computational efficiency.

I. PCA Results Evaluation

Principal Component Analysis (PCA) was applied to the best-performing XGBoost model to evaluate the effect of dimensionality reduction on classification performance. The PCA-transformed model was compared against the original XGBoost classifier using accuracy, precision, recall, and F1-score.

After applying PCA, the model maintained strong recall performance, achieving a testing recall of 87.42%, indicating that the classifier remained effective at detecting diabetic patients. However, precision and F1-score decreased compared to the original XGBoost model. The testing F1-score dropped from 0.8010 to 0.6509 after PCA transformation.

These results indicate that although PCA reduced the dimensionality of the dataset and preserved high sensitivity, some important discriminative information was lost during feature transformation. Consequently, the original XGBoost model without PCA achieved better overall classification performance and remained the final selected model for diabetes prediction.

VI. CONCLUSION

This paper presented a machine learning framework for diabetes disease classification using patient medical and lifestyle

information. Several preprocessing techniques were applied, including duplicate removal, handling inconsistent categorical values, feature scaling, outlier analysis, and data balancing preparation. Exploratory Data Analysis (EDA) was also conducted to better understand feature distributions and relationships with diabetes occurrence.

Multiple machine learning models were implemented and compared, including Logistic Regression, Gaussian Naive Bayes, MLP, AdaBoost, Random Forest, XGBoost, and Support Vector Machine (SVM). Hyperparameter optimization techniques such as GridSearchCV, Randomized Search, and Particle Swarm Optimization (PSO) were used to improve model performance.

Since the dataset was imbalanced, the F1-score was considered the most important evaluation metric because it provides a balanced measure between precision and recall. Experimental results showed that XGBoost achieved the best overall performance with the highest testing F1-score, demonstrating strong capability in identifying diabetic patients while minimizing false predictions.

In addition, learning curve analysis showed that the selected models achieved stable convergence and good generalization performance with minimal overfitting. Principal Component Analysis (PCA) was also applied to the best-performing model to evaluate the effect of dimensionality reduction on classification performance and computational efficiency.

Overall, the results demonstrate that machine learning techniques can effectively support diabetes prediction and assist in early disease detection using healthcare data.

Future work may include testing deep learning architectures, applying advanced feature engineering techniques, and using larger real-world medical datasets to further improve prediction performance and model robustness.

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